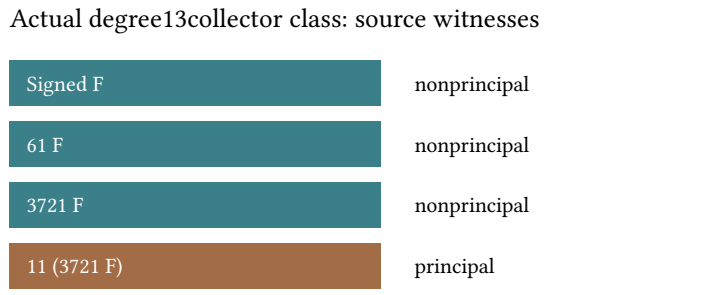
# R47: a collected degree-13 class survives isolation

Exact source/Picard witnesses under retained proper/Riemann–Roch bridges. No independent verdict, selected Koblitz-model image or IC speed claim.




signed47-shift0-01/cofactor47-shift0-01; retained proper/RRbridge; no chosen target image

## Complete inputs from a new collector orbit

All ten points were fixed before section evaluation. Native precision-17 jets pass five source equations, eight units/inverses, all thirteen descent conjugates and a corrupted top-nu control. All 68 original/descended sections agree; 196/324 products are evaluated. All 588 old F32 reference rows match. Evaluation ranks are 68/196/324 in 2210 binary coordinates, not relation ranks.

Each effective divisor is 23Q+Pi, degree 115+13=128. Every one of ten condition matrices has rank 128 and complete small/large kernels 68/196. All thick-Q conditions, residue coordinates, kernel recombinations and full precision-154 global vectors are checked and retained. Completeness uses the existing genus-61 proper/pole/RR bridge, not rank alone.

## Signed class and exact isolation outcomes

The actual ten-step deck order is checked up to coefficient Frobenius. Mixed weights are (4,1,-1,1,-1,-4,-1,1,-1,1). Positive-minus-negative and paired-difference computations agree. Both signed zero tests have witness dimension zero. Every quotient uses all generators; every product has an independent arithmetic check.

The new F and 61F are nonprincipal. The isolated class 3721F is nonprincipal, while 11 times and 121 times that class are principal. Direct 3721 and successive 61,61 calculations agree. Thus the isolated source class has exact order 11 under the retained Picard interpretation. The total order of unisolated F was not determined, and the old order 671 was not assumed.

## Formal boundary and next test

Lean proves common-padding cancellation, equality of signed expressions, and the conditional exact-order implication. It does not import native arrays or formalize the global proper model or Picard witness interpretation. Attempt/audit counts are in results.json; warnings and any failures remain.

Next compute actual diamond action, separate residual characters, compare with old lines, and use this arbitrary collected class to test nonzero action of a proper factor correspondence. New orbit does not imply an independent relation. Chosen-model transfer, saturation/polarization, theta/node187 and complete principal rows remain open. Fixed n=5 carrier, not larger-n transfer.

Sources: Khuri-Makdisi math/0409209v3, R35/R40 complete spaces, R41 class engine and R45/R46 collector; details and limits in LITERATURE.md/REPORT.md.